Dyadic Radon Profiles in the Fabius–Rvachev Web

Spectral reconstruction, generalized Thue–Morse signs, q-sampled cumulants, Pascal factorizations, and Lambert frontiers

A corpus-audited frontier research report

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Based on a recursive review of the current and historically indexed `*.tex` corpus under
`https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs`

Abstract

The Fabius–Rvachev documentation corpus already contains a large network of results on the signed Thue–Morse sequence, exact dyadic Fabius values, q-Pochhammer and Gaussian-binomial formulae, infinite sinc products, Bernoulli and Bell cumulants, Legendre representations, inverse-Fabius Lambert–W asymptotics, and several higher-rank spectral constructions. The present report begins at a boundary not found in the audited sources: a *tomographic classification* of all dyadic linear projections built from independent copies of the classical Rvachev law.

For a nonnegative integer direction profile $c = (c_h)_{h\geq 0}$ satisfying $\sum_h c_h 2^{-h} < \infty$, let

$$Y_c = \sum_{h\geq 0} \sum_{\ell=1}^{c_h} 2^{-h} X_{h,\ell},$$

where the $X_{h,\ell}$ are independent centered Rvachev variables. Its Fourier transform is

$$\Psi_c(t) = \prod_{j\geq 0} \operatorname{sinc}_\pi(t/2^j)^{a_j}, \qquad a_j = \sum_{h\leq j} c_h,$$

and the zero at any integer of 2-adic valuation v has multiplicity

$$m_v = \sum_{h=0}^{v} c_h(v-h+1).$$

The first principal theorem proves that this spectral profile determines the projection uniquely:

$$c_v = m_v - 2m_{v-1} + m_{v-2}.$$

Thus dyadic direction profiles are in bijection with nonnegative integer-valued discretely convex zero-multiplicity sequences. Their generating functions satisfy

$$A(q) = \frac{C(q)}{1-q}, \qquad M(q) = \frac{C(q)}{(1-q)^2}, \qquad \mathcal{Z}_c(s) = \zeta(s)\frac{C(2^{-s})}{1-2^{-s}}.$$

A second group of results shows that every normalized even-cumulant ratio is a q-sample of the same direction polynomial or power series:

$$\frac{\kappa_{2n}(Y_c)}{\kappa_{2n}(X)} = C(4^{-n}).$$

The shifted ratio sequence is therefore a Hausdorff moment sequence carried by the dyadic grid $\{4^{-h}\}$; all alternating finite differences and Hankel minors are nonnegative. Bell-polynomial moments and Legendre coefficients become finite transforms of these q-samples. A third theorem classifies the spectral sign sequence

$$\varepsilon_c(N) = (-1)^{\sum_j (a_j \bmod 2) b_j(N)}:$$

it is 2-automatic exactly when the parity word $(a_j \bmod 2)$ is eventually periodic, and finite blocks have a precisely computable Prouhet cancellation order. A fourth theorem proves the sharp logarithmic Fourier envelope for regularly growing exponents $a_j \sim Aj^d$:

$$\limsup_{x\to\infty} \frac{\log|\Psi_c(x)|}{(\log x)^{d+2}} = -\frac{A}{(d+1)(d+2)(\log 2)^{d+1}}.$$

The existing Pascal–Rvachev hierarchy receives a new geometric factorization: its rank-r law is an explicit infinite Radon projection of classical rank-one Rvachev variables, with scale multiplicities $\binom{h-1}{r-2}$. The report also derives exact positive dyadic-comb cubature for every profile, an inverse-Fabius coordinate representation of the whole projection cone, and a detailed conjectural Lambert–W endpoint program. Fully commented Python code, exact CSV tables, vector and raster figures, the source-audit record, and build instructions accompany the manuscript. Package-wide checksum ledgers are retired; their historical provenance remains recoverable from Git history. The word "new" throughout means apparently new relative to the audited repository corpus, not a claim of worldwide historical priority.

Contents

1 Scope, audit method, and contribution boundary

1.1 The living documentation tree

The requested source is not a single article. At the inspected snapshot it contains a primary Fabius–Rvachev exposition, a mathematical glossary, a non-elementarity paper, a Lean walkthrough, transcribed source papers, a frozen archive, and a semi-formalized frontier system. The latter was reorganized on 28 August 2026 into thematic consolidated volumes. Therefore a filename count from an earlier checkout is not, by itself, a reliable map of current mathematical content.

The audit used three complementary layers.

A1. The current public tree, the current frontier README, the current draft manifest, and the principal consolidated sources were inspected at the level of theorem statements, proof interfaces, open problems, and result registers.

A2. A preserved path-complete ledger from 27 August 2026 was used to retain coverage of paths later absorbed into consolidated manuscripts. It records 78 visible TeX paths at that checkpoint, including generated table fragments; another preserved audit records 57 distinct theorem- bearing artifacts under a different deduplication convention. The discrepancy is explained by consolidation, generated fragments, and frozen revisions, not by ignored source families.

A3. Prior frontier packages in the user's file library were phrase- and formula-searched for every candidate structure developed below. Rediscovered material was removed from the novelty ledger.

The complete historical path ledger is included in the archive as `corpus_inventory_2026-08-27.txt`; the current claim-oriented audit is included as `CORPUS_AUDIT.md`.

1.2 Established material treated as input

The report does *not* relabel the following as new:

- the random-uniform series for the centered Rvachev law and its compactly supported C^∞ density;
- the Fabius/Rvachev functional-differential equations, exact dyadic values, rational moments, and the $q = 1/2$ Gaussian-binomial formulae;
- Bernoulli cumulants, complete Bell-polynomial moments, Appell deconvolution, and polynomial synthesis by shifted up-functions;
- the canonical infinite sinc product, integer zero multiplicities $1 + \nu_2(n)$, arithmetic rays, spectral determinants, and Fourier-decay analyses;
- the Pascal–Rvachev product hierarchy, its existing zero divisor, generalized spectral signs, spectral zeta function, endpoint saddle, and rank asymptotics;
- self-sampling and dyadic-comb quadrature in the classical and previously developed higher-rank settings;
- Legendre, Jacobi, Chebyshev, Lagrange, and repeated-integration representation schemes;
- inverse-Fabius algebraic germs, all-orders Lambert–Bell asymptotics, quantile transforms, and the existing integration/fractional-calculus corpus;
- cyclotomic q-limits, root-of-unity natural boundaries, Stein–Koopman calculus, and free/Boolean cumulant investigations developed in prior packages.

These statements are imported interfaces, marked **[I]**. The new layer is their composition through a general dyadic direction profile.

1.3 Result ledger

Table 1: Principal results and proof status.

ID	Result	Contribution
R1	[N] Profile law	Infinite-dimensional Radon projection, support, smooth density, and cumulative sinc exponents.
R2	[N] Spectral inversion	Exact formula $c_v = \Delta^2 m_v$ and classification by discrete convexity.
R3	[N] Zero-zeta transform	$\mathcal{Z}_c(s) = \zeta(s)C(2^{-s})/(1-2^{-s})$.
R4	[N] Cumulant q-sampling	$\kappa_{2n}(Y_c)/\kappa_{2n}(X) = C(4^{-n})$.
R5	[N] Hausdorff rigidity	Complete monotonicity, Hankel positivity, and uniqueness of the direction profile from cumulants.
R6	[N] Digital sign classification	Automaticity iff the exponent-parity word is eventually periodic; exact Prouhet order.
R7	[N] Sharp Fourier envelope	Exact limsup constant for $a_j \sim Aj^d$.
R8	[N] Pascal factorization	Every Pascal rank is an explicit dyadic projection of classical up-laws.
R9	[N] Profile cubature	Sharp uniform degree for positive dyadic-comb rules from m_H.
R10	[N] Inverse-Fabius coordinates	Exact representation of all profile laws by independent inverse-Fabius variables.
R11	[E] Computation	Exact finite-difference, Prouhet, Hankel, zeta, and high-precision envelope tests.
R12	[C] Frontier program	Spectral rigidity, Lambert transseries, tomography, stability, and matrix-dilation extensions.

Proof-status convention

A theorem labelled **[N]** is proved in this report. A statement labelled **[I]** is taken from the audited Fabius–Rvachev corpus or from a standard classical source and is restated only to fix normalization. Numerical claims are labelled **[E]**; conjectures and research problems are labelled **[C]**. Computation is never used as a substitute for a proof.

2 Normalization and imported rank-one structure

2.1 The centered Rvachev random variable

Define the normalized sinc function

$$\operatorname{sinc}_\pi z := \frac{\sin(\pi z)}{\pi z}, \qquad \operatorname{sinc}_\pi 0 = 1. \tag{2.1}$$

Let $U_0, U_1, \ldots$ be independent random variables uniformly distributed on $[-1/2, 1/2]$, and put

$$X := \sum_{j=0}^{\infty} 2^{-j} U_j. \tag{2.2}$$

The series converges absolutely and $|X| \le 1$. We use up for its density. This is the centered Rvachev up-law in the normalization of the repository. Its characteristic function in the 2π Fourier convention is

$$\Phi(t) := \mathbb{E}e^{2\pi i t X} = \prod_{j=0}^{\infty} \operatorname{sinc}_\pi(t/2^j). \tag{2.3}$$

The density is even, nonnegative, C^∞, supported on $[-1, 1]$, and flat at both endpoints.

2.2 Fabius and inverse-Fabius coordinates

Let $F_{\mathrm{bd}} : [0,1] \to [0,1]$ be the standard Fabius function, with $Q_F = F_{\mathrm{bd}}^{-1}$. With

$$G(x) := \mathbb{P}(X \le x), \qquad -1 \le x \le 1, \tag{2.4}$$

we have the exact coordinate relation

$$G(x) = F_{\mathrm{bd}}\left(\frac{x+1}{2}\right). \tag{2.5}$$

Indeed, differentiating the right-hand side and using the two halves of the Fabius dilation equation gives $G'(x) = \mathrm{up}(x)$, while both sides vanish at $x = -1$.

Proposition 2.1 ([I] Inverse-Fabius quantile coordinate)**.** *If* $V \sim \mathrm{Unif}[0,1]$, *then*

$$X \overset{d}{=} 2Q_F(V) - 1. \tag{2.6}$$

Proof. The quantile of $G(x) = F_{\mathrm{bd}}((x+1)/2)$ is $G^{-1}(v) = 2Q_F(v) - 1$. □

2.3 Bernoulli cumulants

The moment generating function is

$$M_X(z) = \mathbb{E}e^{zX} = \prod_{j=0}^{\infty} \frac{\sinh(z/2^{j+1})}{z/2^{j+1}}. \tag{2.7}$$

Writing $\kappa_n(X)$ for the ordinary cumulants, the repository's Bernoulli–Mersenne formula is

$$\kappa_{2n}(X) = \frac{2^{2n-1}B_{2n}}{n(4^n-1)}, \qquad \kappa_{2n+1}(X) = 0. \tag{2.8}$$

For example,

$$\kappa_2(X) = \frac{1}{9}, \qquad \kappa_4(X) = -\frac{2}{225}, \qquad \kappa_6(X) = \frac{16}{11907}. \tag{2.9}$$

These imported cumulants will serve as spectral probes of the new direction profiles.

3 Dyadic Radon profiles

3.1 Definition and geometric meaning

Definition 3.1 (Dyadic direction profile)**.** A *dyadic direction profile* is a sequence $c = (c_h)_{h\ge 0}$ of nonnegative integers such that

$$S_c := \sum_{h=0}^{\infty} c_h 2^{-h} < \infty. \tag{3.1}$$

For independent copies $X_{h,\ell}$ of X, define

$$Y_c := \sum_{h=0}^{\infty} \sum_{\ell=1}^{c_h} 2^{-h} X_{h,\ell}. \tag{3.2}$$

The deterministic bound in (3.1) makes (3.2) absolutely convergent for every point in the underlying product space. If only finitely many c_h are nonzero, the law is the ordinary Radon projection of the product density $\prod_{h,\ell} \mathrm{up}(x_{h,\ell})$ along the coefficient vector whose entries are 2^{-h}. For infinitely many occupied shells it is the compatible projective limit of those finite-dimensional Radon projections. This is the reason for the terminology.

Define the profile generating function and its cumulative exponents by

$$C(q) := \sum_{h\geq 0} c_h q^h, \qquad a_j := \sum_{h=0}^{j} c_h, \qquad A(q) := \sum_{j\geq 0} a_j q^j. \tag{3.3}$$

The support condition implies convergence of $C(q)$ at least for $|q| < 1/2$. All identities below are also valid formally.

Theorem 3.2 (**[N]** Radon-profile law)**.** *Let c be a dyadic direction profile. Then:*

(i) *Y_c has support exactly $[-S_c, S_c]$.*

(ii) *If $c \neq 0$, its law has a nonnegative C^∞ density g_c with compact support.*

(iii) *Its characteristic function is*

$$\Psi_c(t) := \mathbb{E}e^{2\pi i t Y_c} = \prod_{h\geq 0} \Phi(t/2^h)^{c_h} = \prod_{j\geq 0} \operatorname{sinc}_\pi(t/2^j)^{a_j}. \tag{3.4}$$

(iv) *The generating functions satisfy the first triangular integration law*

$$A(q) = \frac{C(q)}{1-q}. \tag{3.5}$$

Proof. The product of intervals $\prod_{h,\ell}[-1,1]$ is compact, and the linear map $L_c((x_{h,\ell})) = \sum_{h,\ell} 2^{-h} x_{h,\ell}$ is continuous because its series converges uniformly. Its image is the Minkowski sum of intervals $[-2^{-h}, 2^{-h}]$ repeated c_h times, hence the full interval $[-S_c, S_c]$. The product probability measure has full support, so its push-forward has exactly that support.

Choose the first occupied shell h_0 and one of its coordinates. Write $Y_c = 2^{-h_0}X + R$, with R independent. The law of $2^{-h_0}X$ has a compactly supported C^∞ density, and convolving it with the finite measure $\mathcal{L}(R)$ preserves C^∞ regularity and nonnegativity.

Independence gives the first product in (3.4). Substitute (2.3) and collect the factor $\operatorname{sinc}_\pi(t/2^j)$: it occurs once for every outer shell $h \leq j$, with total multiplicity a_j. Finally,

$$A(q) = \sum_{j\geq 0}\sum_{h\leq j} c_h q^j = \sum_{h\geq 0} c_h \frac{q^h}{1-q} = \frac{C(q)}{1-q}.$$

□

Proposition 3.3 (**[N]** Profile addition and renormalization)**.** *For profiles c, d and independent variables,*

$$Y_{c+d} \overset{d}{=} Y_c + Y_d, \qquad \Psi_{c+d} = \Psi_c \Psi_d. \tag{3.6}$$

Let $Tc = (c_{h+1})_{h\geq 0}$ be the left shift. Then

$$Y_c \overset{d}{=} \sum_{\ell=1}^{c_0} X_\ell + \frac{1}{2} Y'_{Tc}, \tag{3.7}$$

where all terms on the right are independent, and

$$C(q) = c_0 + qC_{Tc}(q). \tag{3.8}$$

Proof. Both statements are obtained by partitioning the independent coordinates in (3.2). In the second, reindex $h \mapsto h+1$ in the positive-scale tail. □

Example 3.4 (Classical and polynomial profiles)**.** The classical up-law is $c = (1, 0, 0, \ldots)$, so $a_j = 1$. If $c_h = 1$ for every h, then $a_j = j+1$, $S_c = 2$, and

$$\Psi_c(t) = \prod_{j\geq 0} \operatorname{sinc}_\pi(t/2^j)^{j+1}.$$

More generally, choosing $c_h = (h+1)^d - h^d$ gives $a_j = (j+1)^d$ and produces the full hierarchy of logarithmic Fourier orders treated in section 7.

4 Integer zero profiles and exact spectral inversion

4.1 Multiplicity as a second triangular integration

For $n \in \mathbb{Z} \setminus \{0\}$ write $v = \nu_2(|n|)$. A factor $\operatorname{sinc}_\pi(t/2^j)$ vanishes at $t = n$ exactly when $j \le v$, and then the zero is simple.

Theorem 4.1 ([N] Integer-zero profile)**.** *The zero of Ψ_c at every nonzero integer of 2-adic valuation v has multiplicity*

$$m_v := \operatorname{ord}_{t=n} \Psi_c(t) = \sum_{j=0}^{v} a_j = \sum_{h=0}^{v} c_h(v - h + 1). \tag{4.1}$$

If

$$M(q) := \sum_{v \ge 0} m_v q^v, \tag{4.2}$$

then

$$M(q) = \frac{A(q)}{1-q} = \frac{C(q)}{(1-q)^2}. \tag{4.3}$$

Proof. The first equality counts the vanishing sinc factors in (3.4). Substituting $a_j = \sum_{h \le j} c_h$ and reversing the finite triangular sum gives the second expression. A second application of the calculation in (3.5) gives (4.3). □

4.2 Discrete-convex classification

Set $m_{-1} = m_{-2} = 0$ and use the forward difference $\Delta m_v = m_v - m_{v-1}$.

Theorem 4.2 ([N] Spectral reconstruction and classification)**.** *The direction profile is recovered from the integer-zero multiplicities by*

$$\boxed{c_v = \Delta^2 m_v = m_v - 2m_{v-1} + m_{v-2}.} \tag{4.4}$$

Consequently, a sequence $(m_v)_{v \ge 0}$ occurs as the integer-zero profile of a dyadic Radon projection if and only if

$$\Delta^2 m_v \in \mathbb{Z}_{\ge 0} \quad \text{for every } v, \qquad \sum_{v \ge 0} (\Delta^2 m_v) 2^{-v} < \infty. \tag{4.5}$$

The realizing profile is unique. It has finite support if and only if m_v is eventually affine in v.

Proof. From (4.1),

$$\Delta m_v = a_v, \qquad \Delta^2 m_v = a_v - a_{v-1} = c_v,$$

with the stated zero initial conditions. This proves necessity and uniqueness. Conversely, if (4.5) holds, define $c_v = \Delta^2 m_v$. Twice summing the second differences from the zero initial conditions reconstructs the original m_v and gives (4.1); the weighted condition makes c an admissible profile. Finally, $c_v = 0$ eventually exactly when $\Delta^2 m_v = 0$ eventually, which is equivalent to eventual affinity. □

Remark 4.3 (A monoid isomorphism). Profile addition corresponds to pointwise addition of zero profiles. Thus theorem 4.2 identifies the additive cone of dyadic direction profiles with the additive cone of integer-valued, discretely convex sequences satisfying the support bound. The Fourier zero divisor is not merely an invariant; it is a complete coordinate system for this projection cone.

Example 4.4 (Low-complexity inversions)**.** (a) $m_v = v + 1$ gives $c_0 = 1$ and $c_v = 0$ for $v \ge 1$: the classical law.

(b) $m_v = r(v + 1)$ gives $c_0 = r$ and no other occupied scale: the convolution of r independent up-laws.

(c) $m_v = \binom{v+2}{2}$ gives $c_v = 1$ for all v: one coordinate at every scale.

(d) Any eventually quadratic multiplicity sequence is impossible for a finite-dimensional projection, but may arise from an infinite profile.

4.3 The zero-divisor Dirichlet series

Theorem 4.5 ([N] Direction-to-zeta transform)**.** *For $\Re s$ sufficiently large,*

$$\mathcal{Z}_c(s) := \sum_{n=1}^{\infty} \frac{m_{\nu_2(n)}}{n^s} = \zeta(s)\frac{C(2^{-s})}{1-2^{-s}}. \tag{4.6}$$

Equivalently,

$$\mathcal{Z}_c(s) = (1-2^{-s})\zeta(s)M(2^{-s}). \tag{4.7}$$

Proof. Write $n = 2^v(2k+1)$ and separate the odd part:

$$\begin{aligned}\mathcal{Z}_c(s) &= \sum_{v\geq 0} m_v 2^{-vs} \sum_{k\geq 0}(2k+1)^{-s} \\ &= (1-2^{-s})\zeta(s)M(2^{-s}).\end{aligned}$$

Now substitute (4.3). □

Remark 4.6 (Complex dyadic dimensions). If $C(q)$ extends meromorphically and has a pole of order d at $q=1$, then (4.6) places a lattice of poles of order $d+1$ at $2^{-s}=1$, namely $s = 2\pi ik/\log 2$, subject only to explicit numerator cancellation. Hence the same direction series controls both the integer zero divisor and the familiar dyadic complex-frequency lattice of endpoint and heat-trace analyses.

5 q-sampled cumulants, Hausdorff rigidity, and Bell–Legendre transforms

5.1 Every even cumulant samples the direction series

Cumulants add under independent convolution and scale homogeneously. Combining this elementary fact with (2.8) yields an unexpectedly rigid q-sampling law.

Theorem 5.1 ([N] Cumulant sampling theorem)**.** *For every $n \geq 1$,*

$$\kappa_{2n}(Y_c) = \kappa_{2n}(X)\sum_{h\geq 0} c_h 4^{-nh} = \frac{2^{2n-1}B_{2n}}{n(4^n-1)}C(4^{-n}), \tag{5.1}$$

and all odd cumulants vanish. Thus the normalized ratios

$$R_n(c) := \frac{\kappa_{2n}(Y_c)}{\kappa_{2n}(X)} \tag{5.2}$$

are precisely

$$\boxed{R_n(c) = C(4^{-n}).} \tag{5.3}$$

Proof. Each copy $2^{-h}X$ contributes $2^{-2nh}\kappa_{2n}(X)$ to the $2n$th cumulant. Summing over the c_h independent copies gives (5.1). Symmetry kills all odd cumulants. □

The base-4 sampling is not accidental: cumulants see squares of the dyadic projection coefficients. The values 4^{-n} approach the interior accumulation point 0, so the complete even-cumulant sequence is an analytic tomogram of the direction profile.

5.2 A dyadic Hausdorff moment problem

Define a finite positive measure on $[0,1]$ by

$$\mu_c := \sum_{h\geq 0} c_h 4^{-h}\, \delta_{4^{-h}}. \tag{5.4}$$

It is finite because $\sum_h c_h 4^{-h} \leq \sum_h c_h 2^{-h}$.

Theorem 5.2 ([N] Hausdorff moment and positivity theorem)**.** *The shifted cumulant-ratio sequence is the Hausdorff moment sequence of μ_c:*

$$R_{n+1}(c) = \int_0^1 x^n\, \mathrm{d}\mu_c(x), \qquad n \geq 0. \tag{5.5}$$

Consequently, for every $k, n \geq 0$,

$$(-1)^k \Delta^k R_{n+1}(c) = \sum_{h\geq 0} c_h 4^{-h}(1-4^{-h})^k (4^{-h})^n \geq 0. \tag{5.6}$$

Every Hankel matrix

$$\big(R_{i+j+1}(c)\big)_{0\leq i,j\leq N} \tag{5.7}$$

is positive semidefinite. In particular,

$$R_{n+1}R_{n+3} \geq R_{n+2}^2. \tag{5.8}$$

If the profile has exactly r occupied shells, then the infinite Hankel matrix has rank r.

Proof. Equation (5.5) is the identity

$$\sum_h c_h 4^{-(n+1)h} = \sum_h c_h 4^{-h}(4^{-h})^n.$$

Applying Δ in n multiplies each atom by $4^{-h}-1$; iterating gives (5.6). For any vector $(u_0,\dots,u_N)$,

$$\sum_{i,j=0}^N u_i u_j R_{i+j+1} = \int_0^1 \left(\sum_{i=0}^N u_i x^i\right)^2 \mathrm{d}\mu_c(x) \geq 0,$$

which proves Hankel positivity and (5.8). For finite support, the moment matrix is a Vandermonde–diagonal–Vandermonde transpose factorization, whose rank is the number of distinct atoms. ☐

Corollary 5.3 ([N] Cumulant rigidity)**.** *Two dyadic Radon profiles have the same sequence of even cumulants if and only if they are identical.*

Proof. Equality of even cumulants gives equality of all moments of the finite measures μ_c and μ_d. The Hausdorff moment problem on $[0,1]$ is determinate, so $\mu_c = \mu_d$. The mass at 4^{-h} is $c_h 4^{-h}$, hence $c_h = d_h$ for every h. ☐

Remark 5.4 (An analytic proof)*.* The same conclusion follows from (5.3): the analytic functions C_c and C_d agree on the sequence $4^{-n} \to 0$, so the identity theorem forces equality in a neighborhood of zero and hence coefficientwise. The Hausdorff proof is stronger numerically because it reveals positivity and rank.

5.3 Finite q-Vandermonde reconstruction

Proposition 5.5 ([N] Finite-profile reconstruction)**.** *Assume $c_h = 0$ for $h > H$. Then the first $H+1$ ratios $R_1, \dots, R_{H+1}$ determine $c_0, \dots, c_H$ by the nonsingular system*

$$R_n = \sum_{h=0}^{H} c_h (4^{-n})^h, \qquad 1 \le n \le H+1. \tag{5.9}$$

Its determinant is

$$\prod_{1 \le i < j \le H+1} (4^{-j} - 4^{-i}) \ne 0. \tag{5.10}$$

Proof. This is ordinary polynomial interpolation of $C(q)$ at the distinct geometric nodes $q_n = 4^{-n}$. The determinant is the Vandermonde determinant. □

Remark 5.6 (Connection with the repository's q-analogue program). The nodes form a geometric comb, but the unknowns are not values of a generic function: they are nonnegative integer direction multiplicities. This adds a Hausdorff-moment and lattice-recovery constraint to geometric Lagrange/Newton interpolation. Exact recovery is algebraically simple but increasingly ill conditioned because the nodes cluster doubly exponentially near zero; this stability frontier is developed in section 12.

5.4 Bell-polynomial moments

Let $\mathcal{B}_n^{\exp}(x_1, \dots, x_n)$ denote the complete exponential Bell polynomial. Set

$$K_{2r}(c) := \frac{2^{2r-1} B_{2r}}{r(4^r - 1)} C(4^{-r}), \qquad K_{2r+1}(c) := 0. \tag{5.11}$$

Corollary 5.7 ([N] Bell moment transform)**.** *For every $n \ge 0$,*

$$\mathbb{E} Y_c^n = \mathcal{B}_n^{\exp}\big(K_1(c), \dots, K_n(c)\big). \tag{5.12}$$

In particular, every moment is an explicit polynomial in the finite sample set

$$C(4^{-1}), C(4^{-2}), \dots, C(4^{-\lfloor n/2 \rfloor}).$$

If c has finite support, all moments are rational.

Proof. The complete Bell polynomial converts cumulants to moments. Insert theorem 5.1. Rationality follows because a finite integer profile makes each $C(4^{-r})$ rational. □

5.5 Legendre coefficients as a finite q-Bell transform

Assume for this subsection that $S_c = 1$, so g_c is supported on $[-1, 1]$. Define

$$\Lambda_n(c) := \int_{-1}^{1} P_n(x) g_c(x) \, \mathrm{d}x = \mathbb{E} P_n(Y_c). \tag{5.13}$$

The coefficient of P_n in the Legendre expansion of g_c is $(2n+1)\Lambda_n(c)/2$.

Theorem 5.8 ([N] Legendre–Bell–q formula)**.** *For every $n \ge 0$,*

$$\Lambda_n(c) = 2^{-n} \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k \frac{(2n-2k)!}{k!(n-k)!(n-2k)!} \mathcal{B}_{n-2k}^{\exp}\big(K_1(c), \dots, K_{n-2k}(c)\big). \tag{5.14}$$

All odd coefficients vanish. Thus the complete Legendre spectrum is a triangular finite transform of Bernoulli numbers and the geometric samples $C(4^{-r})$.

Proof. Use the explicit monomial expansion

$$P_n(x) = 2^{-n} \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k \frac{(2n-2k)!}{k!(n-k)!(n-2k)!} x^{n-2k}$$

and substitute (5.12). Symmetry gives the vanishing of odd terms. □

Remark 5.9 (Closing a representation triangle). The repository already relates the up-function to Legendre polynomials and expresses exact moments through Bernoulli/Bell data. Formula (5.14) adds a third vertex: a whole tomographic cone of Rvachev projections is encoded by one direction series C, sampled at $q = 4^{-r}$. The finite transform can therefore be run in either direction: profile → cumulants → Legendre data, or sufficiently rich Legendre moment data → moments → profile.

6 Generalized Thue–Morse signs and exact Prouhet order

6.1 The spectral sign word

Let $b_j(N) \in \{0, 1\}$ be the jth binary digit of N and define

$$\alpha_j := a_j \pmod 2. \tag{6.1}$$

Theorem 6.1 ([N] Sign formula)**.** *For $N \geq 0$ and $t \in (N, N+1)$,*

$$\operatorname{sgn} \Psi_c(t) = \varepsilon_c(N) := (-1)^{\sum_{j\geq 0} a_j \lfloor N/2^j \rfloor} = (-1)^{\sum_{j\geq 0} \alpha_j b_j(N)}. \tag{6.2}$$

For $M \geq 1$, its length-2^M generating polynomial is

$$E_{c,M}(z) := \sum_{N=0}^{2^M-1} \varepsilon_c(N) z^N = \prod_{j=0}^{M-1} \left(1 + (-1)^{\alpha_j} z^{2^j}\right). \tag{6.3}$$

For $|z| < 1$ the infinite generating function converges to

$$E_c(z) = \prod_{j=0}^{\infty} \left(1 + (-1)^{\alpha_j} z^{2^j}\right). \tag{6.4}$$

Proof. The factor $\operatorname{sinc}_\pi(t/2^j)$ crosses one simple zero at every positive multiple of 2^j, so its sign on $(N, N+1)$ is $(-1)^{\lfloor N/2^j \rfloor}$. Multiplying with exponent a_j gives the first formula. Modulo two, $\lfloor N/2^j \rfloor \equiv b_j(N)$, proving the digit formula.

Every $N < 2^M$ has a unique binary expansion. Expanding the product in (6.3) and choosing either 1 or the monomial from each factor yields precisely the coefficient $(-1)^{\sum_j \alpha_j b_j(N)}$. Normal convergence for $|z| < 1$ gives the infinite product. □

The classical profile has $\alpha_j = 1$ for every j, so (6.3) reduces to the finite Thue–Morse product $\prod_{j<M}(1 - z^{2^j})$.

6.2 Automaticity is exactly eventual parity periodicity

Theorem 6.2 ([N] Automaticity classification)**.** *The sign sequence $(\varepsilon_c(N))_{N\geq 0}$ is 2-automatic if and only if the parity word $(\alpha_j)_{j\geq 0}$ is eventually periodic.*

Proof. Suppose first that α_j is eventually periodic. For every $k \geq 0$ and $0 \leq r < 2^k$, split the binary digits into the fixed low block of r and the shifted digits of N. Then

$$\varepsilon_c(2^k N + r) = \sigma_{k,r} (-1)^{\sum_{j\geq 0} \alpha_{j+k} b_j(N)}, \qquad \sigma_{k,r} \in \{\pm 1\}.$$

An eventually periodic word has only finitely many distinct tails, so these subsequences take only finitely many forms. The 2-kernel is finite, and ε_c is 2-automatic.

Conversely, consider the subsequences

$$\varepsilon_c(2^k N) = (-1)^{\sum_{j\geq 0} \alpha_{j+k} b_j(N)}.$$

They belong to the 2-kernel of ε_c. If two tails $(\alpha_{j+k})_{j\geq 0}$ and $(\alpha_{j+\ell})_{j\geq 0}$ differ at index r, then the corresponding subsequences differ at $N = 2^r$. Thus a finite 2-kernel implies that only finitely many tails occur, which is equivalent to eventual periodicity. □

Corollary 6.3 ([N] Mahler equation)**.** *If α_j is periodic from the beginning with period P, then*

$$E_c(z) = \left[\prod_{j=0}^{P-1} \left(1 + (-1)^{\alpha_j} z^{2^j}\right)\right] E_c(z^{2^P}). \tag{6.5}$$

An eventually periodic parity word gives the same equation after extracting a finite prefix factor and rescaling the tail by a power of two.

Proof. Split the infinite product (6.4) into residue classes of the index modulo P and shift the tail. □

6.3 Prouhet cancellation

Let

$$d_M(c) := \#\{0 \leq j < M : \alpha_j = 1\}. \tag{6.6}$$

Theorem 6.4 ([N] Exact Prouhet order)**.** *The polynomial $E_{c,M}$ has a zero of exact order $d_M(c)$ at $z = 1$. Consequently,*

$$\sum_{N=0}^{2^M-1} \varepsilon_c(N) N^k = 0, \qquad 0 \leq k < d_M(c). \tag{6.7}$$

The first nonzero falling-factorial moment occurs at degree $d_M(c)$.

Proof. In (6.3), a factor with $\alpha_j = 1$ is $1 - z^{2^j}$ and has a simple zero at 1; a factor with $\alpha_j = 0$ is $1 + z^{2^j}$ and is nonzero there. This proves the exact order. The first d_M ordinary derivatives vanish at 1, which annihilates the corresponding falling-factorial moments. Ordinary powers are triangular combinations of falling factorials, proving (6.7). □

Example 6.5 (Automatic and nonautomatic profiles)**.** (a) Classical: $c = (1, 0, \ldots)$ gives $\alpha_j = 1$ and $d_M = M$, the maximal Thue–Morse cancellation.

(b) $c_0 = c_1 = 1$ gives $a_0 = 1$ and $a_j = 2$ for $j \geq 1$; hence $\varepsilon_c(N) = (-1)^{b_0(N)}$ is periodic and $d_M = 1$.

(c) The Pascal profile in section 8 has $\alpha_j = \binom{j}{r-1} \bmod 2$, periodic by Lucas' theorem.

(d) If $c_h = 1$ at $h = 0, 1, 2, 4, 8, 16, \ldots$ and is zero elsewhere, the cumulative parity toggles at powers of two and is not eventually periodic. The resulting spectral sign sequence is therefore not 2-automatic, despite being generated by a completely explicit sparse dyadic projection.

7 A sharp logarithmic Fourier envelope

The repository contains detailed rank-one and Pascal-rank Fourier-decay analyses. The direction-profile formalism gives a single theorem covering all polynomially growing exponent sequences, with a sharp coefficient that can be read directly from a_j.

Lemma 7.1 (Weighted triangular asymptotic). *For an integer $d \geq 0$,*

$$\sum_{j=0}^{H}(j+1)^d(H-j) = \frac{H^{d+2}}{(d+1)(d+2)} + \mathcal{O}_d(H^{d+1}). \tag{7.1}$$

Proof. After dividing by H^{d+2}, the sum is a Riemann sum for $\int_0^1 u^d(1-u)\,\mathrm{d}u = 1/((d+1)(d+2))$. Euler–Maclaurin or elementary power-sum estimates give the stated error. □

Theorem 7.2 ([N] Sharp profile Fourier envelope). *Assume that for some integer $d \geq 0$ and $A > 0$,*

$$a_j = A(j+1)^d + \mathcal{O}((j+1)^{d-1}), \tag{7.2}$$

where for $d = 0$ the condition means $a_j = A + o(1)$. Then

$$\boxed{\limsup_{x\to\infty} \frac{\log|\Psi_c(x)|}{(\log x)^{d+2}} = -\frac{A}{(d+1)(d+2)(\log 2)^{d+1}}.} \tag{7.3}$$

The limsup is attained along the rational dyadic ray

$$x_H = \frac{2^H}{3}. \tag{7.4}$$

Proof. Put $L = \log 2$ and $H = \lfloor \log_2 x \rfloor$. For $0 \leq j \leq H$ we have $x/2^j \geq 1$ and

$$|\operatorname{sinc}_\pi(x/2^j)| \leq \frac{2^j}{\pi x}.$$

Discarding the tail factors, whose moduli are at most one, gives

$$\begin{aligned}
\log|\Psi_c(x)| &\leq -\sum_{j=0}^{H} a_j(\log x - jL) + \mathcal{O}\left(\sum_{j=0}^{H} a_j\right) \\
&= -AL\sum_{j=0}^{H}(j+1)^d(H-j) + \mathcal{O}_d(H^{d+1}) \\
&= -\frac{AL}{(d+1)(d+2)}H^{d+2} + \mathcal{O}_d(H^{d+1}).
\end{aligned}$$

Since $\log x = HL + \mathcal{O}(1)$, this proves the upper bound in (7.3).

For $x_H = 2^H/3$ and $j \leq H$, put $k = H - j$. Because $\left|\sin(\pi 2^k/3)\right| = \sqrt{3}/2$,

$$\log\left|\operatorname{sinc}_\pi(2^k/3)\right| = -kL + \log\frac{3\sqrt{3}}{2\pi}. \tag{7.5}$$

Hence the head of the product is

$$-AL\sum_{j=0}^{H}(j+1)^d(H-j) + \mathcal{O}_d(H^{d+1}).$$

For the tail $j > H$, use $\log\operatorname{sinc}_\pi y = -\pi^2 y^2/6 + \mathcal{O}(y^4)$ near zero. Since $a_j = \mathcal{O}(j^d)$,

$$\sum_{j>H} a_j \left|\log\operatorname{sinc}_\pi(2^{H-j}/3)\right| = \mathcal{O}_d\left(\sum_{r\geq 1}(H+r)^d 4^{-r}\right) = \mathcal{O}_d(H^d).$$

Thus the upper main term is attained along (7.4), and division by $(\log x_H)^{d+2}$ proves the equality. □

Corollary 7.3 (Growth-degree dictionary). *The exponent-growth degree d raises the logarithmic Fourier-decay order to $d+2$. In particular:*

profile behavior	*exponent behavior*	*leading log decay*
$c=(1,0,\dots)$	$a_j=1$	$-(2\log 2)^{-1}(\log x)^2$
$c_h\equiv 1$	$a_j=j+1$	$-(6(\log 2)^2)^{-1}(\log x)^3$
$c_h=2h+1$	$a_j=(j+1)^2$	$-(12(\log 2)^3)^{-1}(\log x)^4$
$a_j\sim Aj^d$	*regular degree* d	*theorem* 7.2

Remark 7.4 (The same triangular kernel appears twice). The zero multiplicity is the discrete triangular convolution $m_v=\sum_h c_h(v-h+1)$. The Fourier exponent in theorem 7.2 is another triangular convolution, $\sum_j a_j(H-j)$. Spectral multiplicity and high-frequency decay are therefore two successive integrations of the same direction profile. This explains why a second difference reconstructs the geometry while two cumulative sums control the logarithmic envelope.

8 The Pascal hierarchy as a projection of classical up-laws

8.1 Imported Pascal products

For $r\ge 1$, define the Pascal–Rvachev product

$$\Phi_r(t):=\prod_{j=0}^{\infty}\operatorname{sinc}_\pi(t/2^j)^{\binom{j}{r-1}}, \tag{8.1}$$

with the convention $\binom{j}{r-1}=0$ when $j<r-1$. The audited corpus already develops its support, smooth density, refinement ladder, zero multiplicity

$$\operatorname{ord}_{t=n}\Phi_r(t)=\binom{\nu_2(|n|)+1}{r}, \tag{8.2}$$

and many arithmetic and asymptotic consequences. The following theorem gives a new rank-one geometric realization.

Theorem 8.1 ([N] Pascal projection factorization). *Let X_r have characteristic function Φ_r. For $r=1$, $X_1=X$. For $r\ge 2$,*

$$\boxed{X_r\overset{d}{=}\sum_{h=r-1}^{\infty}\sum_{\ell=1}^{\binom{h-1}{r-2}}2^{-h}X_{h,\ell},} \tag{8.3}$$

where the $X_{h,\ell}$ are independent classical centered Rvachev variables. Equivalently, the Pascal direction profile is

$$c_h^{(r)}=\begin{cases}\binom{h-1}{r-2}, & h\ge r-1,\\ 0, & h<r-1.\end{cases} \tag{8.4}$$

Proof. The cumulative exponent of (8.4) is, by the hockey-stick identity,

$$\sum_{h=0}^{j}c_h^{(r)}=\sum_{h=r-1}^{j}\binom{h-1}{r-2}=\binom{j}{r-1}.$$

The characteristic function in theorem 3.2 is therefore exactly (8.1). Characteristic functions determine probability laws. □

Corollary 8.2 ([N] Exact scale norms and support)**.** *For $p > 0$ and $r \geq 1$,*

$$\sum_{h\geq 0} c_h^{(r)} 2^{-ph} = (2^p - 1)^{-(r-1)}. \tag{8.5}$$

In particular, every rank has support $[-1, 1]$, and

$$\kappa_{2n}(X_r) = \frac{\kappa_{2n}(X)}{(4^n - 1)^{r-1}} = \frac{2^{2n-1} B_{2n}}{n(4^n - 1)^r}. \tag{8.6}$$

Proof. For $r \geq 2$, the negative-binomial generating function gives

$$\sum_{h\geq r-1} \binom{h-1}{r-2} x^h = \frac{x^{r-1}}{(1-x)^{r-1}}.$$

Set $x = 2^{-p}$ to obtain (8.5); the case $r = 1$ is immediate. At $p = 1$ the sum is one, giving the support. At $p = 2n$, apply theorem 5.1. ☐

Corollary 8.3 (Pascal zero profile and sign mask)**.** *The factorization recovers*

$$m_v^{(r)} = \binom{v+1}{r} \tag{8.7}$$

and identifies the sign mask as

$$\alpha_j^{(r)} = \binom{j}{r-1} \pmod 2. \tag{8.8}$$

By Lucas' theorem this mask is periodic with period a power of two, so the Pascal spectral sign is 2-automatic. Its finite-block Prouhet order is

$$d_{r,M} = \#\left\{0 \leq j < M : \binom{j}{r-1} \text{ is odd}\right\}. \tag{8.9}$$

Proof. The zero multiplicity is the hockey-stick sum of the exponents, or follows directly from theorem 4.1. The sign statements follow from theorems 6.2 and 6.4 and Lucas' theorem. ☐

Remark 8.4 (Geometric meaning of rank)*.* The rank parameter does not require a new primitive atom. It counts how many classical up-coordinates are placed on each dyadic shell. The binomial shell multiplicity is the discrete derivative of the Pascal exponent. This turns the existing algebraic refinement ladder into a concrete tomography of a single rank-one law.

9 Positive dyadic-comb cubature and profile Appell polynomials

The corpus already contains classical self-sampling and higher-rank comb results. Here the zero-profile classification supplies the exact degree for *every* dyadic Radon profile without repeating a rank-by-rank argument. Use the Fourier transform

$$\widehat{f}_{2\pi}(\xi) = \int_{\mathbb{R}} f(x) \mathrm{e}^{-2\pi \mathrm{i} \xi x} \, \mathrm{d}x. \tag{9.1}$$

Since g_c is even, $\widehat{g_c}_{2\pi} = \Psi_c$.

Theorem 9.1 ([N] Uniform profile cubature)**.** *Fix $H \geq 0$, put $Q = 2^H$, and assume $m_H \geq 1$. For every phase $\theta \in \mathbb{R}$ and every polynomial p of degree strictly less than m_H,*

$$\boxed{\int_{\mathbb{R}} p(x) g_c(x) \, \mathrm{d}x = \frac{1}{Q} \sum_{k\in\mathbb{Z}} p\left(\frac{k+\theta}{Q}\right) g_c\left(\frac{k+\theta}{Q}\right).} \tag{9.2}$$

The sum is finite and all weights are nonnegative. The degree threshold is sharp in the following uniform sense: the identity for degree m_H cannot hold for every phase θ.

Proof. For $p(x) = x^j$, Poisson summation on the shifted lattice gives

$$\frac{1}{Q}\sum_{k\in\mathbb{Z}} p\left(\frac{k+\theta}{Q}\right) g_c\left(\frac{k+\theta}{Q}\right) = \sum_{n\in\mathbb{Z}} \mathrm{e}^{2\pi i n\theta}\, \widehat{x^j g_c}_{2\pi}(Qn). \tag{9.3}$$

Up to a nonzero constant, $\widehat{x^j g_c}_{2\pi}(\xi)$ is the jth derivative of $\Psi_c(\xi)$. For $n \neq 0$,

$$\mathrm{ord}_{\xi=Qn}\,\Psi_c = m_{H+\nu_2(|n|)} \geq m_H.$$

Hence all nonconstant Fourier modes vanish when $j < m_H$. The constant mode is $\int x^j g_c(x)\,\mathrm{d}x$. Linearity gives (9.2). Compact support makes the sum finite; $g_c \geq 0$ gives positivity.

For $j = m_H$, the Fourier coefficient at $n = 1$ is proportional to $\Psi_c^{(m_H)}(Q)$, which is nonzero because the zero at Q has exact order m_H. Therefore the right-hand side is not constant as a function of θ, proving uniform sharpness. □

Corollary 9.2 (Zero profile as polynomial precision)**.** *At lattice scale 2^{-H} the positive comb precision is exactly one less than the spectral multiplicity m_H. Conversely, measuring the sharp precisions at every dyadic lattice reconstructs the direction profile by*

$$c_H = (d_H + 1) - 2(d_{H-1} + 1) + (d_{H-2} + 1), \tag{9.4}$$

where $d_H = m_H - 1$ is the uniform exactness degree and the negative-index terms are interpreted as zero.

9.1 Profile Appell deconvolution

Let

$$M_c(z) := \mathbb{E}\mathrm{e}^{zY_c} \tag{9.5}$$

be the profile moment generating function. Define the monic Appell sequence $\mathcal{A}_{c,n}$ by

$$\frac{\mathrm{e}^{xz}}{M_c(z)} = \sum_{n=0}^{\infty} \mathcal{A}_{c,n}(x)\frac{z^n}{n!}. \tag{9.6}$$

Its coefficients are complete Bell polynomials in the negatives of the cumulants (5.1).

Proposition 9.3 ([N] Profile deconvolution)**.** *For every $n \geq 0$,*

$$\mathbb{E}\,\mathcal{A}_{c,n}(x + Y_c) = x^n. \tag{9.7}$$

If $n < m_H$, then for every phase θ,

$$x^n = \frac{1}{Q}\sum_{k\in\mathbb{Z}} \mathcal{A}_{c,n}\left(x + \frac{k+\theta}{Q}\right) g_c\left(\frac{k+\theta}{Q}\right). \tag{9.8}$$

Proof. Average the generating function (9.6) after replacing x by $x + Y_c$; the factor $M_c(z)$ cancels. Coefficient comparison gives (9.7). Apply theorem 9.1 to the polynomial $y \mapsto \mathcal{A}_{c,n}(x + y)$. □

Remark 9.4 (Separation from prior self-sampling results). Poisson summation and Appell deconvolution are established tools in the repository. The contribution here is the universal precision coordinate m_H, its inversion to c_H, and the consequent classification of all profile cubature ladders. Classical and Pascal rules are specializations.

10 Inverse-Fabius coordinates and the Lambert endpoint frontier

10.1 Exact quantile realization of the projection cone

Let $V_{h,\ell}$ be independent uniform variables on $[0,1]$. Combining proposition 2.1 with the profile definition gives the following exact nonlinear coordinate representation.

Theorem 10.1 ([N] Inverse-Fabius Radon representation)**.** *For every dyadic direction profile,*

$$\boxed{Y_c \overset{d}{=} \sum_{h\geq 0}\sum_{\ell=1}^{c_h} 2^{-h}\big(2Q_F(V_{h,\ell})-1\big).} \tag{10.1}$$

Writing $D_c := S_c + Y_c \in [0, 2S_c]$, one has

$$D_c \overset{d}{=} 2\sum_{h\geq 0}\sum_{\ell=1}^{c_h} 2^{-h} Q_F(V_{h,\ell}), \tag{10.2}$$

and the right-endpoint deficit is

$$S_c - Y_c \overset{d}{=} 2\sum_{h\geq 0}\sum_{\ell=1}^{c_h} 2^{-h}\big(1-Q_F(V_{h,\ell})\big). \tag{10.3}$$

Proof. Replace each independent $X_{h,\ell}$ in (3.2) by the equal-in-law quantile variable from (2.6). Adding S_c or subtracting from S_c gives the last two identities. □

Thus every linear tomographic law is also a nonlinear sum of inverse-Fabius coordinates. This exact bridge is useful because the repository's sharpest endpoint information is naturally written for $Q_F(y)$ and for the lower branch W_{-1} of Lambert's function.

10.2 Laplace factorization

Let $D := 1 + X = 2Q_F(V)$ and define

$$L_1(s) := \mathbb{E}\mathrm{e}^{-sD} = \int_0^1 \mathrm{e}^{-2sI(v)}\,\mathrm{d}v. \tag{10.4}$$

Proposition 10.2 ([N] Endpoint Laplace profile)**.** *For $s \geq 0$,*

$$L_c(s) := \mathbb{E}\mathrm{e}^{-sD_c} = \prod_{h\geq 0} L_1(2^{-h}s)^{c_h}. \tag{10.5}$$

Equivalently,

$$-\log L_c(s) = \sum_{h\geq 0} c_h\,[-\log L_1(2^{-h}s)]. \tag{10.6}$$

Proof. Use the independent sum representation (10.2) and multiply the Laplace transforms. □

The same profile that is twice integrated in the zero multiplicities and sampled at 4^{-n} by cumulants now acts as a dyadic Mellin convolution on the rank-one endpoint Laplace exponent. This is the correct interface for a uniform Lambert theory.

10.3 A conjectural all-orders endpoint law

The existing rank-one asymptotics have a quadratic logarithmic leading term, a lower-Lambert coordinate, and a bounded periodic phase. The existing Pascal hierarchy raises the polynomial degree of the endpoint saddle. The general profile theorem suggests that the exponent-growth degree, rather than the special binomial form, is the true invariant.

Conjecture 10.3 ([C] Profile Lambert transseries)**.** *Assume $a_j = A(j+1)^d + \mathcal{O}((j+1)^{d-1})$ with $A > 0$. There exist a positive constant γ_c, a degree-$(d+2)$ polynomial P_c, and bounded periodic coefficient functions such that, as $x \downarrow 0$, the endpoint probability*

$$H_c(x) := \mathbb{P}(S_c - Y_c \le x) \tag{10.7}$$

has an all-orders transseries organized by

$$\Lambda_c(x) := -(d+1)W_{-1}\left(-\gamma_c x^{1/(d+1)}\right), \tag{10.8}$$

with leading structure

$$-\log H_c(x) = P_c(\Lambda_c(x)) + \Lambda_c(x)^d \Omega_c(\log_2 \Lambda_c(x)) + \mathcal{O}\left(\Lambda_c(x)^{d-1} \log \Lambda_c(x)\right). \tag{10.9}$$

The leading coefficient of P_c depends only on A and d; lower coefficients depend on the asymptotic jet of a_j or, equivalently, the singular expansion of $C(q)$ at $q = 1$.

Remark 10.4. For $d = 0$, (10.8) reduces to the classical rank-one lower-Lambert scale and (10.9) has a quadratic leading polynomial. For the Pascal rank r, $d = r - 1$ and the coordinate becomes the known rank-r form $-rW_{-1}(-\gamma x^{1/r})$. The conjecture therefore interpolates established endpoints without asserting that their lower-order periodic functions are universal.

11 Exact and high-precision experiments

11.1 Reproducibility boundary

The accompanying program `numerical_experiments.py` uses exact integer or rational arithmetic for:

- profile integration and second-difference inversion;
- Pascal shell multiplicities;
- digit-sign identities and Prouhet power sums;
- cumulant-ratio finite differences and Hankel determinants.

High-precision `mpmath` arithmetic is used only for the zero-zeta convergence table and the logarithmic Fourier products. The latter are computed as sums of logarithms; no underflowed direct product is used. The plots are presentation devices, not proof evidence.

11.2 Profile inversion

For the finite profile

$$c = (1, 2, 1, 0, 3, 0, 0, \ldots), \tag{11.1}$$

the successive transforms begin

h	0	1	2	3	4	5	6
c_h	1	2	1	0	3	0	0
a_h	1	3	4	4	7	7	7
m_h	1	4	8	12	19	26	33
$\Delta^2 m_h$	1	2	1	0	3	0	0

The program verifies the inversion for four qualitatively different profile families through valuation 18.

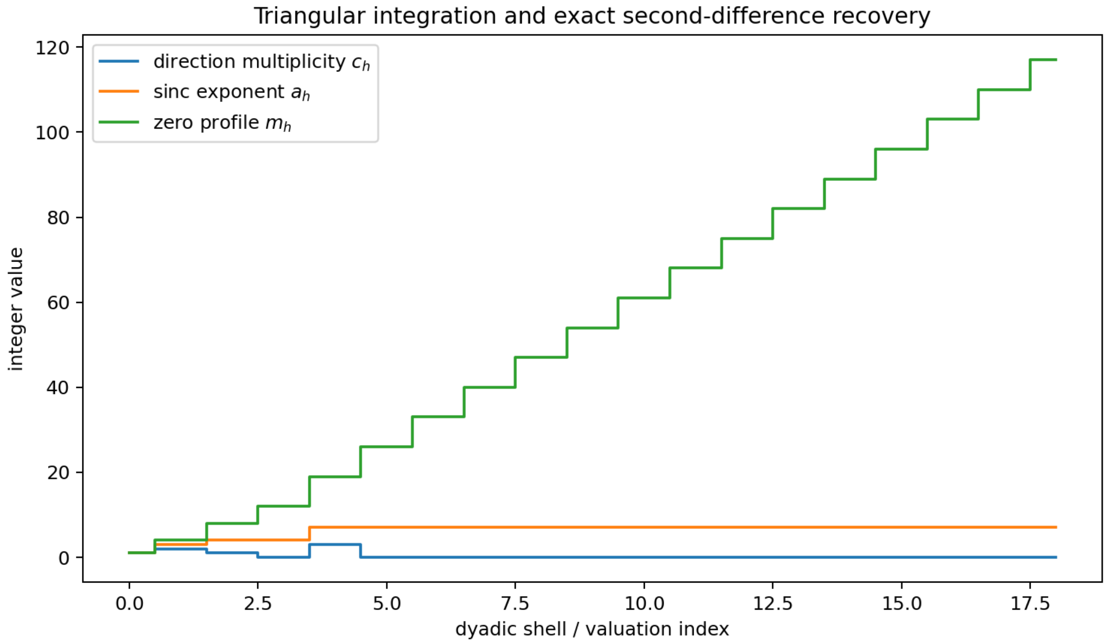

Figure 1: **[E]** A direction profile is integrated once to the sinc exponents and twice to the integer-zero multiplicities. The second difference recovers the original profile exactly.

11.3 Digital signs and Prouhet cancellation

For four profiles, the sign computed from binary digits was compared with the parity of all crossed sinc zeros for every $0 \le N < 1024$; all values agreed. The script then checked 28 finite blocks. In each block all power sums below the order $d_M(c)$ vanished exactly. For the classical profile, the first nonzero sums at block sizes $8, 16, 32$ are respectively $-48, 1536, -122880$, in agreement with the exact Thue–Morse product.

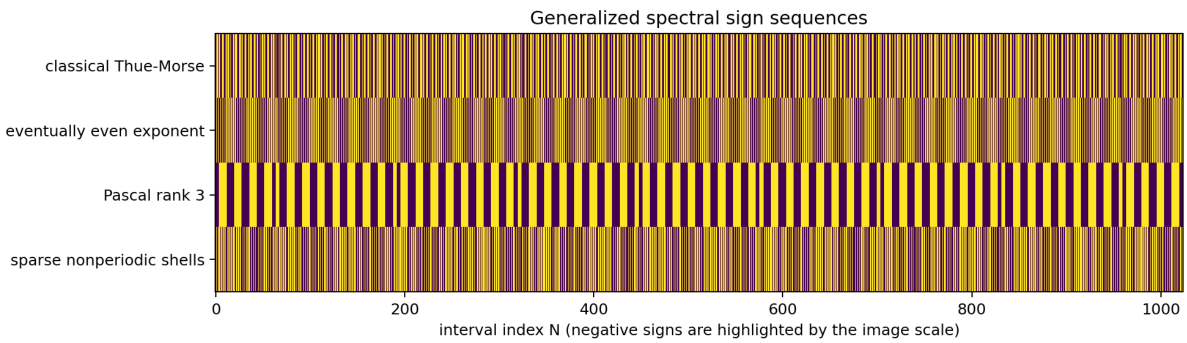

Figure 2: **[E]** Prefixes of four generalized spectral sign sequences. The third row is a Pascal mask; the fourth comes from sparse power-of-two shells and is provably nonautomatic by theorem 6.2.

11.4 Cumulant-ratio moments

For the finite profile

$$c = (2, 1, 3, 1, 2, 2, 1), \tag{11.2}$$

the first ratios are

$$R_1 = \frac{10089}{4096}, \quad R_2 = \frac{34804257}{16777216}, \quad R_3 = \frac{138563297409}{68719476736}. \tag{11.3}$$

All signed finite differences through order eight were nonnegative. Exact leading Hankel determinants through size six were strictly positive, as predicted by the seven occupied dyadic atoms.

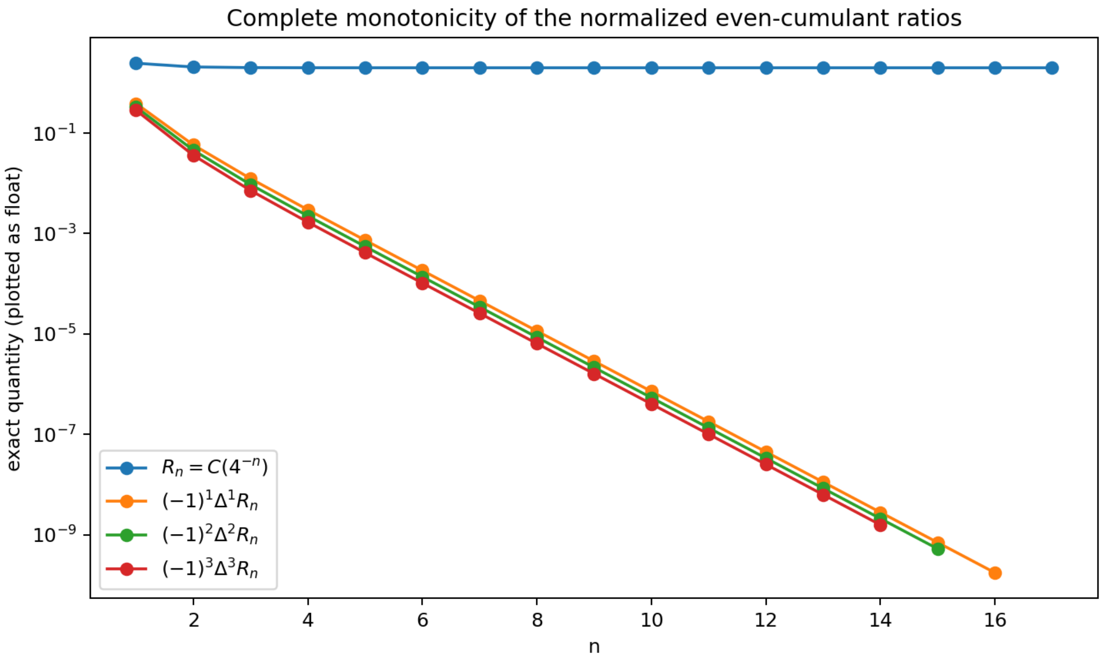

Figure 3: **[E]** Complete monotonicity of the exact cumulant-ratio sequence. The plotted floating values come from exact rational data.

11.5 Zero zeta and Fourier envelope

At $s = 3.75$ for the profile $c = (1, 2, 0, 3, 1)$, the direct sum through $N = 500000$ agrees with (4.6) to an absolute error below 3.8×10^{-16}.

For the exponent families $a_j = (j + 1)^d$, $0 \le d \le 3$, the sharp ray $x_H = 2^H/3$ was evaluated through $H = 500$. The terminal ratios are:

d	H	observed ratio	predicted constant
0	500	-0.7281923692	-0.7213475204
1	500	-0.3528989083	-0.3468948302
2	500	-0.2560241321	-0.2502317256
3	500	-0.2228908502	-0.2166048418

The slowly decaying H^{-1} correction is visible, but all four curves move toward their proved constants.

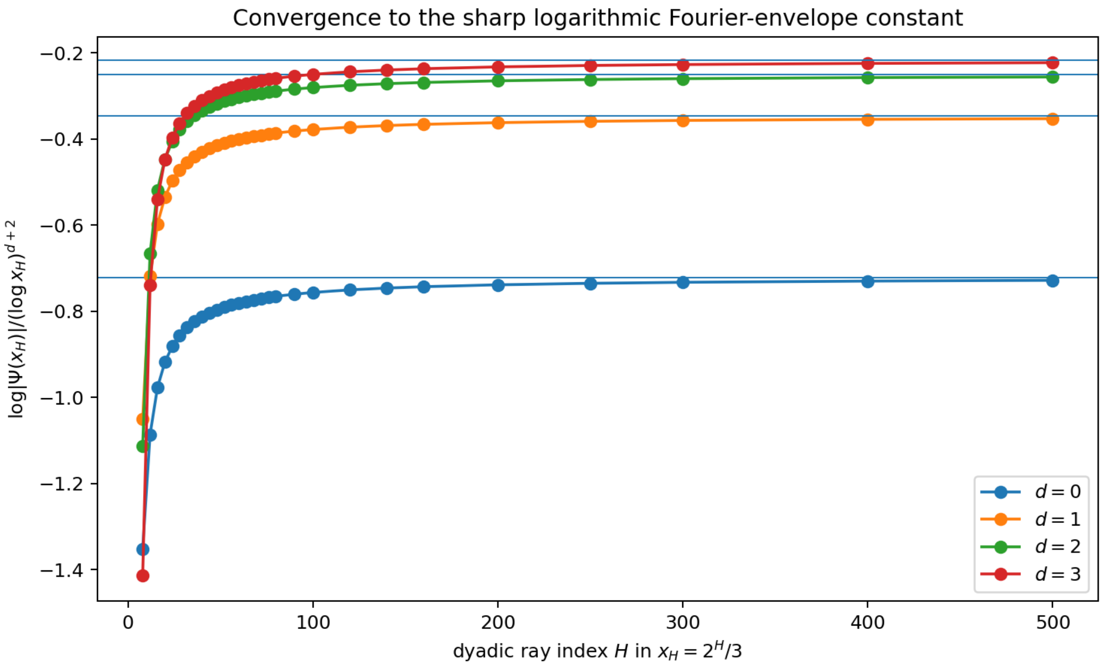

Figure 4: **[E]** Convergence along the sharp rational dyadic ray. The horizontal lines are the constants in (7.3).

12 Conjectures and a frontier research program

12.1 Spectral rigidity beyond the dyadic model

Conjecture 12.1 (**[C]** Dyadic spectral rigidity)**.** *Let*

$$\Theta(t) = \prod_{\lambda\in\Lambda} \Phi(\lambda t)^{r_\lambda}$$

be the characteristic function of a compactly supported projection of independent up-laws, with positive coefficients λ and integer multiplicities r_λ. Suppose every nonzero integer is a zero and its multiplicity depends only on $\nu_2(n)$. After a common rescaling, every λ is a dyadic power and $\Theta = \Psi_c$ for a unique profile c.

A possible proof should compare the smallest non-dyadic denominator in Λ with the exact integer zero set. The obstruction is that different scaled sinc zero lattices can overlap accidentally. A divisor-theoretic argument for entire functions of exponential type may isolate the dyadic lattice.

12.2 Stable q-tomography

Exact recovery in proposition 5.5 is ill conditioned. The nodes 4^{-n} cluster at zero, so high shells contribute only beyond very small scales.

Problem 12.2 (Regularized profile recovery)**.** *Develop error bounds for recovering a sparse or monotone integer profile from noisy ratios $R_1, \dots, R_N$. Compare:*

(i) *constrained geometric Vandermonde inversion;*

(ii) *Prony recovery of the dyadic Hausdorff measure μ_c;*

(iii) *total-variation or ℓ^1 regularization on the shell multiplicities;*

(iv) *modular reconstruction using the integrality of c_h.*

The objective is a theorem giving a recoverable shell depth as a function of relative cumulant precision.

12.3 Tomography from several directions

The present profile uses coefficients from the one-dimensional set $\{2^{-h}\}$. A product of several Fabius–Rvachev fields admits many linear projections.

Problem 12.3 (Finite-angle Fabius tomography)**.** *Given a finite collection of dyadic projection directions, characterize when the corresponding zero profiles determine the full multiset of coordinate weights. Seek analogues of discrete tomography uniqueness, switching components, and compressed-sensing guarantees, with the integer zero divisor as the measurement vector.*

12.4 Base-b and matrix-dilation profiles

For an integer $b \geq 2$, replace 2^{-h} by b^{-h} and the rank-one atom by a base-b compactly supported refinable density. The zero multiplicity then samples b-adic valuations, while the sign sequence becomes a digit-weighted b-kernel object.

Conjecture 12.4 (**[C]** Base-b automaticity criterion)**.** *Under a simple zero-lattice hypothesis, the spectral sign sequence of a base-b profile is b-automatic exactly when the shell-exponent residue word modulo two is eventually periodic. For odd b, the correct sign alphabet may require more than two phases and should be expressed through roots of unity.*

A deeper extension replaces scalar dilation by an expanding integer matrix. The direction generating function becomes multivariate, valuations are replaced by divisibility along lattice orbits, and one expects a matrix Mahler system rather than a scalar equation.

12.5 Optimization of smoothness, support, and cubature precision

At fixed support budget $S_c = \sum_h c_h 2^{-h}$, the precision at level H is

$$m_H = \sum_{h=0}^{H} c_h (H - h + 1). \tag{12.1}$$

Placing mass at coarse scales is expensive in support but contributes to many future levels; fine-scale mass is cheap but arrives late.

Problem 12.5 (Profile design)**.** *Solve the integer optimization problems that maximize one or more of:*

(i) *m_H at a prescribed lattice level;*

(ii) *the asymptotic Fourier order $d + 2$;*

(iii) *a weighted sum of exactness degrees over several levels;*

(iv) *conditioning of the positive cubature weights;*

subject to support, variance, or coordinate-count budgets. Determine whether Pascal profiles are extremal for any natural convex criterion.

12.6 Lambert phase tomography

Conjecture 10.3 predicts that the singular jet of $C(q)$ at $q = 1$ controls the endpoint polynomial, while the dyadic complex poles control periodic corrections. A practical program is:

1. derive the Mellin transform of (10.6);

2. express its poles directly through $C(2^{-s})$;

3. obtain the large-s Laplace exponent by residue summation;

4. solve the saddle equation in the coordinate (10.8);

5. transfer the result to the inverse-Fabius deficit sum (10.3).

This would unify the rank-one and Pascal endpoint theories without assuming a binomial exponent sequence.

12.7 Arithmetic and exact-value questions

Finite profiles have rational cumulants and moments, but pointwise values of g_c at dyadic rationals involve convolutions of up-values and need not inherit the simplest rank-one denominator formulas.

Problem 12.6 (Dyadic arithmetic of profile densities)**.** *For finite profiles, determine:*

(i) *whether every dyadic value of g_c is rational;*

(ii) *exact denominator valuations in terms of c and the point denominator;*

(iii) *q-binomial formulae for the convolution weights;*

(iv) *finite Thue–Morse representations compatible with* (6.3).

The Pascal factorization is a particularly structured test case.

13 Formalization roadmap

The main algebraic results are suitable for staged Lean formalization.

13.1 Tier I: finite combinatorics

1. triangular sums $a_j = \sum_{h \le j} c_h$ and $m_v = \sum_{j \le v} a_j$;

2. second-difference inversion and eventual-affinity criterion;

3. finite sign product (6.3);

4. exact Prouhet order at $z = 1$;

5. Pascal hockey-stick factorization.

These require no analysis beyond finite sums and polynomial multiplicities.

13.2 Tier II: probability and generating functions

1. construction of Y_c under the deterministic support bound;

2. characteristic-function rearrangement;

3. cumulant scaling and $C(4^{-n})$ sampling;

4. the atomic Hausdorff measure and Hankel positivity;

5. inverse-Fabius quantile transport.

13.3 Tier III: complex and Fourier analysis

1. exact zero orders of the infinite product;
2. Poisson summation for compactly supported C^∞ functions;
3. uniform cubature sharpness;
4. the global Fourier upper bound and the rational-ray lower bound;
5. meromorphic continuation consequences of (4.6).

13.4 Tier IV: conjectural endpoint analysis

Formalization should follow, not precede, an analytic proof of the profile Lambert transseries. The likely reusable components are Mellin inversion, residue bookkeeping on a vertical pole lattice, explicit estimates for W_{-1}, and composition of periodic asymptotic scales.

14 Conclusion

A dyadic Radon profile turns many apparently separate parts of the Fabius–Rvachev theory into different transforms of one nonnegative integer sequence:

observable	transform of the profile
sinc exponent	$a_j = \sum_{h\le j} c_h$
integer-zero multiplicity	$m_v = \sum_{h\le v} c_h(v-h+1)$
direction recovery	$c_v = \Delta^2 m_v$
zero zeta	$\zeta(s)C(2^{-s})/(1-2^{-s})$
even cumulant ratio	$C(4^{-n})$
Hausdorff measure	$\sum_h c_h 4^{-h}\delta_{4^{-h}}$
spectral sign	binary digit pairing with $a_j \bmod 2$
Prouhet order	number of odd exponents in a finite prefix
Fourier envelope	two cumulative integrations of the growth of c_h
endpoint Laplace law	$\prod_h L_1(2^{-h}s)^{c_h}$

The complete integer zero divisor determines the geometry; the even cumulants provide a second, independent q-tomographic reconstruction; digital parity produces generalized Thue–Morse sequences; and polynomial exponent growth determines a sharp logarithmic Fourier constant. The Pascal hierarchy is revealed as an explicit arrangement of classical rank-one up-laws rather than an unrelated family of atoms.

The next decisive step is an all-orders endpoint theorem for general profiles. Such a theorem would make $C(q)$ the common source of the zero-zeta pole lattice, the Fourier envelope, the inverse-Fabius deficit, and the nested Lambert phase. The finite-difference classification proved here isolates the right parameter space for that program.

A Proof supplements

A.1 Support and smoothness in the infinite product model

The support argument in theorem 3.2 can be phrased entirely in measure-theoretic terms. Let

$$K_c = \prod_{h\ge 0}\prod_{\ell=1}^{c_h}[-1,1]$$

with the product topology. The profile bound makes $L_c(x) = \sum_{h,\ell} 2^{-h} x_{h,\ell}$ a uniformly convergent series of continuous functions, hence continuous. For any $y \in [-S_c, S_c]$, the convex set K_c contains points mapped to $-S_c$ and S_c, and its continuous linear image is convex, hence the whole interval. Because the product up-measure has support K_c, the push-forward support is $L_c(K_c)$.

For smoothness, if h_0 is occupied, remove one coordinate at that scale. If $\rho_{h_0}(x) = 2^{h_0}\,\mathrm{up}(2^{h_0}x)$, then $g_c = \rho_{h_0} * \nu$ for a compactly supported probability measure ν. Every derivative is $g_c^{(r)} = \rho_{h_0}^{(r)} * \nu$, proving global C^∞ regularity and compact support without differentiating an infinite product.

A.2 Exact first nonzero Prouhet derivative

Let $J_M = \{j < M : \alpha_j = 1\}$. Factoring at $z = 1$ gives

$$E_{c,M}(z) = \prod_{j \in J_M} (1 - z^{2^j}) \prod_{j \notin J_M} (1 + z^{2^j}).$$

Therefore

$$E_{c,M}^{(d_M)}(1) = d_M!\,(-1)^{d_M} 2^{\sum_{j \in J_M} j} 2^{M-d_M}, \tag{A.1}$$

which is nonzero. This gives an exact falling-factorial moment at the first surviving degree and can be converted to an ordinary-power moment with Stirling numbers.

A.3 A direct complete-monotonicity identity

With $\Delta R_n = R_{n+1} - R_n$,

$$\begin{aligned}
(-1)^k \Delta^k R_{n+1} &= (-1)^k \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} R_{n+j+1} \\
&= \sum_h c_h 4^{-(n+1)h} \sum_{j=0}^{k} (-1)^j \binom{k}{j} 4^{-jh} \\
&= \sum_h c_h 4^{-(n+1)h} (1 - 4^{-h})^k.
\end{aligned}$$

This is (5.6). Equality in a nontrivial Hankel minor occurs precisely when the test polynomial vanishes on every occupied atom of μ_c.

B Selected exact data

B.1 Pascal direction profiles

$r \backslash h$	0	1	2	3	4	5	6	7
1	1	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1
3	0	0	1	2	3	4	5	6
4	0	0	0	1	3	6	10	15
5	0	0	0	0	1	4	10	20
6	0	0	0	0	0	1	5	15

Cumulative summation down each row produces $\binom{h}{r-1}$, the exponent in (8.1).

B.2 First cumulant-ratio inequalities

For every profile,

$$\begin{aligned} R_2 &\le R_1, \\ R_3 - 2R_2 + R_1 &\ge 0, \\ R_1R_3 - R_2^2 &\ge 0, \\ \det\begin{pmatrix} R_1 & R_2 & R_3 \\ R_2 & R_3 & R_4 \\ R_3 & R_4 & R_5 \end{pmatrix} &\ge 0. \end{aligned}$$

These are necessary conditions for any proposed even-cumulant sequence to come from a dyadic Radon profile. They are not sufficient unless the representing Hausdorff measure is additionally supported on the exact grid $\{4^{-h} : h \ge 0\}$ with masses divisible by 4^{-h} in the integer sense specified by (5.4).

C Corpus and reproducibility manifest

C.1 Source families screened

Source family	Material screened for overlap
Primary exposition and glossary	normalization, random series, functional equations, exact values, moments, cumulants, inverse, notation
Lean walkthrough and theorem index	formalized Fabius/up interfaces, dyadic arithmetic, Bell/Appell and Fourier declarations
Canonical frontier synthesis	current theorem registers, conjectures, provenance, and absorbed historical drafts
Thue–Morse volumes	finite products, Prouhet cancellation, automatic and Mahler formulae, rarefaction, digital sums
Exponent and q-series volumes	q-Pochhammer, Gaussian binomials, geometric laws, inverse q-functions, cyclotomic limits
Spectra and arithmetic	zero multiplicities, determinants, heat traces, arithmetic rays, Pascal hierarchy, Fourier envelopes
Integration and transforms	Mellin/Laplace/Fourier/Cauchy/Stieltjes transforms, fractional calculus, quantile integrals
Inverse and sampling	inverse germs, Lambert–Bell expansions, self-sampling, comb quadrature, Appell reconstruction
Representation frontiers	Legendre/Jacobi/Chebyshev expansions, polynomial synthesis, multiresolution, operator models
Archived papers and revisions	classical Fabius/Rvachev sources, historical variants, frozen theorem formulations
Prior generated packages	Stein–Koopman, free/Boolean, hyperbolic-gamma, cyclotomic q, Pascal, and geometric-comb reports

Exact and close-variant searches were made for: Radon projection, direction profile, zero-profile inversion, second-difference multiplicity recovery, cumulant tomography, dyadic Hausdorff measure, automaticity iff parity periodicity, inverse-Fabius projection, and Pascal factorization into classical up-laws. No direct treatment of the theorem package above was found in the audited snapshot. This is a repository-relative absence statement, not a global-priority certification.

C.2 Archive contents

Path	Purpose
dyadic_radon_profiles_fabius_rvachev.tex	complete LaTeX manuscript
dyadic_radon_profiles_fabius_rvachev.pdf	compiled and visually inspected PDF
numerical_experiments.py	fully commented exact and high-precision experiment driver
data/*.csv	deterministic exact/numerical tables
data/fourier_envelope_table.tex	generated manuscript table
figures/*.pdf	retained vector source figures
figures/*.png	raster figures used in the report
numerical_results.txt	human-readable computation summary
CORPUS_AUDIT.md	current claim-oriented nonduplication audit
corpus_inventory_2026-08-27.txt	preserved recursive TeX path ledger
README.txt	build, reproduction, and file map
requirements.txt	Python dependencies

References

[1] V. Reshetnikov, *ProveIt: Analysis/FabiusFunction documentation corpus*, GitHub repository, github.com/VladimirReshetnikov/ProveIt, accessed 30 August 2026.

[2] J. Fabius, A probabilistic example of a nowhere analytic C^∞-function, *Zeitschrift fuer Wahrscheinlichkeitstheorie und Verwandte Gebiete* **5** (1966), 173–174.

[3] V. A. Rvachev, Compactly supported solutions of functional-differential equations and their applications, *Russian Mathematical Surveys* **45** (1990), no. 1, 87–120.

[4] J. Arias de Reyna, An infinitely differentiable function with compact support: definition and properties, *Revista de la Real Academia de Ciencias Exactas, Fisicas y Naturales de Madrid* **76** (1982), 21–38; English translation, https://arxiv.org/abs/1702.05442.

[5] J. Arias de Reyna, Arithmetic of the Fabius function, *INTEGERS* **18** (2018), Article A51; https://arxiv.org/abs/1702.06487.

[6] J.-P. Allouche and J. Shallit, *Automatic Sequences: Theory, Applications, Generalizations*, Cambridge University Press, 2003.

[7] L. Comtet, *Advanced Combinatorics*, D. Reidel, 1974.

[8] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey, and D. E. Knuth, On the Lambert W function, *Advances in Computational Mathematics* **5** (1996), 329–359.

[9] S. Helgason, *The Radon Transform*, second edition, Birkhauser, 1999.

[10] J. A. Shohat and J. D. Tamarkin, *The Problem of Moments*, American Mathematical Society, 1943.

[11] G. Szego, *Orthogonal Polynomials*, fourth edition, American Mathematical Society, 1975.

[12] E. C. Titchmarsh, *Introduction to the Theory of Fourier Integrals*, second edition, Oxford University Press, 1948.